

Best Programming Practice

1. All values as variables including Fixed, User Inputs, and Results
2. Proper naming conventions for all variables
3. Proper Program Name and Class Name
4. Proper Method Name which indicates action taking inputs and providing result

Sample Program 1: Create a program to find the sum of all the digits of a number given by a user using an array and display the sum.

- a. Use Math.random() and get a 4-digit random integer number
- b. Write a method to count digits in the number
- c. Write a method to return an array of digits from a given number.
- d. Write a method to Find the sum of the digits of the number in the array
- e. Finally, display the sum of the digits of the number

Java

```
// Create SumOfDigit Class to compute the sum of 4 digits random number
class SumOfDigits {
    // Get a 4 digit random number
    public int get4DigitRandomNumber() {
        return (int) (Math.random() * 9000) + 1000;
    }

    // Find the count of digits in the number
    public int countDigits(int number) {
        int count = 0, temp = number;
        while (temp > 0) {
            count++;
            temp /= 10;
        }
        return count;
    }

    // Store the digits of the number in an array
    public int[] getDigits(int number, int count) {
        int[] digits = new int[count];
        int temp = number;
        for (int i = count - 1; i >= 0; i--) {
            digits[i] = temp % 10;
            temp /= 10;
        }
        return digits;
    }
}
```

```
// Find the sum of the elements in an array
public int sumArray(int[] array) {
    int sum = 0;
    for (int i = 0; i < array.length; i++) {
        sum += array[i];
    }
    return sum;
}

public static void main(String[] args) {
    // Get 4 digit random integer number
    SumOfDigits sumOfDigits = new SumOfDigits();
    int number = sumOfDigits.get4DigitRandomNumber();
    System.out.println("The Random Number is: " + number);

    // Get the count of digits
    int count = sumOfDigits.countDigits(number);
    System.out.println("The count of digit is: " + count);

    // Get the array of digits from the number
    int[] digits = sumOfDigits.getDigits(number, count);

    // Find the sum of the digits of the number
    int sum = sumOfDigits.sumArray(digits);

    // Display the sum of the digits of the number
    System.out.println("\nSum of Digits: " + sum);
}
}
```

Level 2 Practice Programs

1. Create a program to find the factors of a number taken as user input, store the factors in an array and display the factors. Also find the sum, sum of square of factors and product of the factors and display the results

Hint =>

- a. Take the input for a number
- b. Write a **static** Method to find the factors of the number and save them in an array and return the array.
- c. To find factors and save to array will have two loops. The first loop to find the count and initialize the array with the count. And the second loop save the factors into the array
- d. Write a method to find the sum of the factors using factors array
- e. Write a method to find the product of the factors using factors array
- f. Write a method to find the sum of square of the factors using **Math.pow()** method

```
import java.lang.Math;
import java.util.Scanner;
public class FactorCalculator {

    public static int[] factor(int n) {
        //creating fact array for storing the factors
        int[] fact = new int[10];
        //creating variable max to indicate maximum length of array
        int max = 10;

        //creating variable k as pointer of array fact
        int k = 0;

        //iterating loop from 1 to number
        for(int i=1; i<=n; i++) {

            //checking whether i is factor of n or not
            if(n%i ==0) {

                //increasing the size of fact if max length reached
                max +=10;
                int[] temp = new int[max];
                for( int j=0; j<=k; j++) {
                    temp[j] = fact[j];
                }
                fact = temp;
            }
        }
    }
}
```

```

        //inserting the factor in array
        fact[k] = i;
        //incrementing the pointer
        k++;
    }

}

//removing extra indexes
int[] temp = new int[k];
for(int i=0; i<k; i++) {
    temp[i] = fact[i];
}
fact=temp;

//returning factors
return fact;
}

//method to find sum of factors
public static int sumOfFactor(int[] fact) {

    //creating variable sum to indicate the sum of factors
    int sum = 0;
    //calculating the sum of factors
    for(int i=0; i<fact.length; i++) {
        sum += fact[i];
    }

    //returning the sum of factors
    return sum;
}

//method to find product of factors
public static int productOfFactor(int[] fact) {

    //creating variable product to indicate the product of factors
    int product = 1;
    //calculating the sum of factors

```

```

        for(int i=0; i<fact.length; i++) {
            product *= fact[i];
        }

        //returning the product of factors
        return product;
    }

    //method to find sum Of Sq of factors
    public static int sumOfSqOfFactor(int[] fact) {

        //creating variable sum of squares to indicate the sum of
        factors

        int sumOfSq = 0;
        //calculating the sum of squares of factors
        for(int i=0; i<fact.length; i++) {
            sumOfSq += Math.pow(fact[i], 2);
        }

        //returning the sum of factors
        return sumOfSq;
    }

    public static void main(String[] args) {

        //creating Scanner object for taking input
        Scanner input = new Scanner(System.in);

        //creating fact array for storing the factors
        int[] fact;

        /*creating variables n, sum, product and sumOfSq to indicate
        number, sum of factors, product of factors and Sum of squares
        of factors respectively*/
        int n, sum, product, sumOfSq;

        //taking user input for a number
        System.out.print("Enter a number : ");
        n = input.nextInt();

        //calling the factor method to find factors
    }

```

```

fact = factor(n);

//calling the sumOfFactor method to find sum of factors
sum = sumOfFactor(fact);

//calling the productOfFactor method to find product of factors
product = productOfFactor(fact);

//calling the sumOfSqOfFactor method to find sum of square of
factors
sumOfSq = sumOfSqOfFactor(fact);

//printing results
System.out.println("The number is " + n);
System.out.print("Factors of number are " );
for(int i=0; i<fact.length; i++) {
    System.out.print(fact[i] + " ");
}
System.out.println("\nThe sum of factors is " + sum);
System.out.println("The product of factors is " + product);
System.out.println("The sum of squares of factors is " +
sumOfSq);

//closing the input object
input.close();
}
}

```

2. Write a program to find the sum of n natural numbers using recursive method and compare the result with the formulae $n*(n+1)/2$ and show the result from both computations is correct.

Hint =>

- a. Take the user input number and check whether it's a Natural number, if not exit
- b. Write a Method to find the sum of n natural numbers using **recursion**
- c. Write a Method to find the sum of n natural numbers using the formulae $n*(n+1)/2$
- d. Compare the two results and print the result

```

import java.util.Scanner;
public class NaturalNumberSumComparison {

```

```
//method to calculate sum recursively
public static int recursiveSum(int n) {

    //calculating the sum
    if(n == 1) {
        return 1;
    } else {
        return (n + recursiveSum(n-1));
    }
}

//method to calculate sum by formula
public static int sumByFormula(int n) {

    // Calculate the sum using the formula
    int sum = n * (n + 1) / 2;

    //returning the sum
    return sum;
}

public static void main(String[] args) {

    // Creating a Scanner object to take user input
    Scanner input = new Scanner(System.in);

    //creating variables
    int number, sumRecursive, sumFormula;

    // Take user input for the number
    System.out.print("Enter a natural number: ");
    number = input.nextInt();

    // Checking if the number is a natural number (greater than 0)
    if (number > 0) {

        // Calculating the sum using recursive function
        sumRecursive = recursiveSum(number);
```

```
//Calculating the sum by formula
sumFormula = sumByFormula(number);

// Printing the results
System.out.println("The sum of " + number + " natural
numbers using the formula is " + sumFormula);
System.out.println("The sum of " + number + " natural
numbers using recursion " + sumRecursive);

// Comparing the results
if (sumFormula == sumRecursive) {
    System.out.println("Both computations are
correct.");
} else {
    System.out.println("There is a difference between
the two computations.");
}
}
else {
    System.out.println("The number " + number + " is not a
natural number.");
}
// Close the scanner
input.close();
}
}
```

3. Write a program that takes a year as input and outputs the Year is a Leap Year or not

Hint =>

- The LeapYear program only works for year ≥ 1582 , corresponding to a year in the Gregorian calendar.
- Also Leap year is divisible by 4 and not divisible by 100 or divisible by 400
- Write a method to check for Leap Year using the conditions a and b

```
import java.util.Scanner;
public class LeapYearByMethod {

    //method to check leap year
    public static void checkLeapYear(int year) {
```



```
// Checking if the year is valid (Gregorian calendar starts
from 1582)
if (year >= 1582) {

    // Determine if the year is a Leap Year using a single if
condition with logical operators
    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 ==
0)) {

        System.out.println("The year " + year + " is a Leap
Year.");
    } else {
        System.out.println("The year " + year + " is not a
Leap Year.");
    }
} else {
    System.out.println("The year " + year + " is not valid.
Please enter a year >= 1582.");
}
}

public static void main(String[] args) {

    // Creating a Scanner object to take user input
    Scanner input = new Scanner(System.in);

    // Taking user input for the year
    System.out.print("Enter a year (>= 1582): ");
    int year = input.nextInt();

    //Checking the given year is leap year or not
    checkLeapYear(year);

    // Closing the scanner
    input.close();
}
}
```

4. Extend or Create a **UnitConverter** utility class similar to the one shown in the notes to do the following. Please define **static** methods for all the UnitConverter class methods. E.g.

public static double convertKmToMiles(double km) =>

- Method To convert kilometers to miles and return the value. Use the following code
`double km2miles = 0.621371;`
- Method to convert miles to kilometers and return the value. Use the following code
`double miles2km = 1.60934;`
- Method to convert meters to feet and return the value. Use the following code to convert
`double meters2feet = 3.28084;`
- Method to convert feet to meters and return the value. Use the following code to convert
`double feet2meters = 0.3048;`

```
import java.util.Scanner;
public class UtilityConventor {

    // Conversion factor for kilometers to miles
    private static final double KM_TO_MILES = 0.621371;

    // Conversion factor for miles to kilometers
    private static final double MILES_TO_KM = 1.60934;

    // Conversion factor for meters to feet
    private static final double METERS_TO_FEET = 3.28084;

    // Conversion factor for feet to meters
    private static final double FEET_TO_METERS = 0.3048;

    //method to convert kilometers to miles
    public static double kilometerToMile(double kilometer) {
        return (kilometer * KM_TO_MILES);
    }

    public static double mileToKilometer(double mile) {
        return (mile * MILES_TO_KM);
    }

    //method to convert meter to feet
    public static double meterToFeet(double meter) {
        return (meter * METERS_TO_FEET);
    }

    //method to convert feet to meter
    public static double feetToMeter(double feet) {
        return (feet * FEET_TO_METERS);
    }

    public static void main(String[] args) {

        //creating Scanner object for taking inputs
        Scanner input = new Scanner(System.in);

        /*creating variables distance, choice and result to indicate distance entered
        by user, choice of user and respective distance after conversion
```

```

respectively*/
    double distance, result;
    int choice;

    //taking user input
    System.out.print("Enter your value : ");
    distance = input.nextDouble();

    // asking the user to select the desired operation to perform and taking user
input
    System.out.print("Enter your choice \n1 to convert kilometer to mile \n2 to
convert miles to kilometer \n3 to convert meter to feet \n4 to convert feet to meter \n");
    choice = input.nextInt();

    //calling the respective method to perform desired operation
    switch (choice) {
        case 1 : {

            //calling the method kilometeToMile to convert kilometer to
mile
            result = kilometerToMile(distance);

            //printing the results
            System.out.print("The distance in kilometer is " + distance + "
and in mile is " + result);
            break;
        }
        case 2 : {

            //calling the method mileToMile to convert kilometer to mile
            result = mileToKilometer(distance);

            //printing the results
            System.out.print("The distance in mile is " + distance + " and
in kilometer is " + result);
            break;
        }
        case 3 : {

            //calling the metmileToFeet to convert meter to feet
            result = meterToFeet(distance);

            //printing the results
            System.out.print("The distance in meter is " + distance + " and
in feet is " + result);
            break;
        }
        case 4 : {

            //calling the methodfeetToMeter to convert feet to meter

```

```

        result = feetToMeter(distance);

        //printing the results
        System.out.print("The distance in feet is " + distance + " and
in meter is " + result);
        break;
    }
    default : {
        System.out.print("You have entered a wrong choice");
    }
}

//closing the input object
input.close();
}
}

```

5. Extend or Create a **UnitConvertor** utility class similar to the one shown in the notes to do the following. Please define **static** methods for all the UnitConvertor class methods. E.g.

public static double convertYardsToFeet(double yards) =>

- Method to convert yards to feet and return the value. Use following code to convert
`double yards2feet = 3;`
- Method to convert feet to yards and return the value. Use following code to convert
`double feet2yards = 0.333333;`
- Method to convert meters to inches and return the value. Use following code to convert
`double meters2inches = 39.3701;`
- Method to convert inches to meters and return the value. Use following code to convert
`double inches2meters = 0.0254;`
- Method to convert inches to centimeters and return the value. Use the following code
`double inches2cm = 2.54;`

```

import java.util.Scanner;
public class UtilityConvertor2 {

    // Conversion factor for yard to feet
    private static final double YARD_TO_FEET = 3;

    // Conversion factor for feet to yard
    private static final double FEET_TO_YARD = 0.333333;

    // Conversion factor for meters to inches
    private static final double METERS_TO_INCHES = 39.3701;

    // Conversion factor for inches to meters
    private static final double INCHES_TO_METERS = 0.0254;
}

```

```
// Conversion factor for inches to centimeter
private static final double INCHES_TO_CM = 2.50;

//method to convert yard to feet
public static double yardToFeet(double yard) {
    return (yard * YARD_TO_FEET);
}

//method to convert feet to yard
public static double feetToYard(double feet) {
    return (feet * FEET_TO_YARD);
}

//method to convert meter to inches
public static double meterToInches(double meter) {
    return (meter * METERS_TO_INCHES);
}

//method to convert inches to meter
public static double inchesToMeter(double inches) {
    return (inches * INCHES_TO_METERS);
}

//method to convert inches to centimeter
public static double inchesToCM(double inches) {
    return (inches * INCHES_TO_CM);
}

public static void main(String[] args) {

    //creating Scanner object for taking inputs
    Scanner input = new Scanner(System.in);

    /*creating variables distance, choice and result to indicate distance entered
    by user, choice of user and respective distance after conversion
    respectively*/
    double distance, result;
    int choice;

    //taking user input
    System.out.print("Enter your value : ");
    distance = input.nextDouble();

    // asking the user to select the desired operation to perform and taking user
    input

    System.out.print("Enter your choice \n1 to convert yard to feet \n2 to
    convert feet to yard \n3 to convert meter to inches \n4 to convert inches to meter \n5 to
    convert inches to centimeter \n");
    choice = input.nextInt();
}
```

```
//calling the respective method to perform desired operation
switch (choice) {
    case 1 : {

        //calling the method yardToFeet to convert yard to feet
        result = yardToFeet(distance);

        //printing the results
        System.out.print("The distance in yard is " + distance + " and
in feet is " + result);
        break;
    }
    case 2 : {

        //calling the method feetToYard to convert feet to yard
        result = feetToYard(distance);

        //printing the results
        System.out.print("The distance in feet is " + distance + " and
in yard is " + result);
        break;
    }
    case 3 : {

        //calling the method meterToInches to convert meter to inches
        result = meterToInches(distance);

        //printing the results
        System.out.print("The distance in meter is " + distance + " and
in inch is " + result);
        break;
    }
    case 4 : {

        //calling the method inchesToMeter to convert inches to meter
        result = inchesToMeter(distance);

        //printing the results
        System.out.print("The distance in inches is " + distance + "
and in meter is " + result);
        break;
    }
    case 5 : {

        //calling the method inchesToCM to convert inches to centimeter
        result = inchesToCM(distance);

        //printing the results
        System.out.print("The distance in inches is " + distance + "
and in centimeter is " + result);
    }
}
```

```

        break;
    }
    default : {
        System.out.print("You have entered a wrong choice");
    }
}

//closing the input object
input.close();
}
}

```

6. Extend or Create a **UnitConvertor** utility class similar to the one shown in the notes to do the following. Please define **static** methods for all the UnitConvertor class methods. E.g.

public static double convertFarhenheitToCelsius(double farhenheit) =>

- Method to convert Fahrenheit to Celsius and return the value. Use the following code
`double farhenheit2celsius = (farhenheit - 32) * 5 / 9;`
- Method to convert Celsius to Fahrenheit and return the value. Use the following code
`double celsius2farhenheit = (celsius * 9 / 5) + 32;`
- Method to convert pounds to kilograms and return the value. Use the following code
`double pounds2kilograms = 0.453592;`
- Method to convert kilograms to pounds and return the value. Use the following code
`double kilograms2pounds = 2.20462;`
- Method to convert gallons to liters and return the value. Use following code to convert
`double gallons2liters = 3.78541;`
- Method to convert liters to gallons and return the value. Use following code to convert
`double liters2gallons = 0.264172;`

```

import java.util.Scanner;
public class UtilityConvertor3 {

    // Conversion factor for pound to kilogram
    private static final double POUND_TO_KILOGRAM = 0.453592;

    // Conversion factor for kilogram to pound
    private static final double KILOGRAM_TO_POUND = 2.20462;

    // Conversion factor for gallon to liter
    private static final double GALLON_TO_LITER = 3.78541;

    // Conversion factor for liter to gallon
    private static final double LITER_TO_GALLON = 0.264172;

    //method to convert farhenheit to celsius
    public static double farhenheitToCelsius(double farhenheit) {

```

```
//creating celsius variable to indicate temperature inn celsius
double celsius = (farhenheit - 32) * 5 / 9;

//returning the temperature in celsius
return celsius;
}

//method to convert celsius to farhenheit
public static double celsiusToFarhenheit(double celsius) {

    //creating celsius variable to indicate temperature in celsius
    double farhenheit = (celsius * 9 / 5) + 32;

    //returning the temperature in farhenheit
    return farhenheit;
}

//method to convert pound to kilogram
public static double poundToKilogram(double pound) {
    return (pound * POUND_TO_KILOGRAM);
}

//method to convert kilogram to pound
public static double kilogramToPound(double kilogram) {
    return (kilogram * KILOGRAM_TO_POUND);
}

//method to convert gallon to liter
public static double gallonToLiter(double gallon) {
    return (gallon * GALLON_TO_LITER);
}

//method to convert liter to gallon
public static double literToGallon(double liter) {
    return (liter * LITER_TO_GALLON);
}

public static void main(String[] args) {

    //creating Scanner object for taking inputs
    Scanner input = new Scanner(System.in);

    /*creating variables distance, choice and result to indicate distance entered
    by user, choice of user and respective distance after conversion
    respectively*/
    double initialUnit, resultUnit;
    int choice;
```



```

        // asking the user to select the desired operation to perform and taking user
input
        System.out.print("Enter your choice \n1 to convert farhenheit to celsius \n2
to convert celsius to farhenheit \n3 to convert pound to kilogram \n4 to convert kilogram to
pound \n5 to convert gallon to liter \n6 to convert liter to gallon \n");
        choice = input.nextInt();

        //calling the respective method to perform desired operation
        switch (choice) {
            case 1 : {

                //taking user input
                System.out.print("Enter temperature in farhenheit : ");
                initialUnit = input.nextDouble();

                //calling the method farhenheitToCelsius to convert farhenheit
to celsius

                resultUnit = farhenheitToCelsius(initialUnit);

                //printing the results
                System.out.print("The temperature in farhenheit is " +
initialUnit + " and in celsius is " + resultUnit);
                break;
            }
            case 2 : {

                //taking user input
                System.out.print("Enter temperature in celsius : ");
                initialUnit = input.nextDouble();

                //calling the method celsiusToFarhenheit to convert celsius to
farhenheit

                resultUnit = celsiusToFarhenheit(initialUnit);

                //printing the results
                System.out.print("The temperature in celsius is " + initialUnit
+ " and in farhenheit is " + resultUnit);
                break;
            }
            case 3 : {

                //taking user input
                System.out.print("Enter weight in pound : ");
                initialUnit = input.nextDouble();

                //calling the method poundToKilogram to convert pound to
kilogram

                resultUnit = poundToKilogram(initialUnit);

                //printing the results

```

```

        System.out.print("The weight in pound is " + initialUnit + "
and in kilogram is " + resultUnit);
        break;
    }
    case 4 : {

        //taking user input
        System.out.print("Enter weight in kilogram : ");
        initialUnit = input.nextDouble();

        //calling the method kilogramToPound to convert kilogram to
pound

        resultUnit = kilogramToPound(initialUnit);

        //printing the results
        System.out.print("The weight in kilogram is " + initialUnit + "
and in pound is " + resultUnit);
        break;
    }
    case 5 : {

        //taking user input
        System.out.print("Enter volume in gallon : ");
        initialUnit = input.nextDouble();

        //calling the method gallonToLiter to convert gallon to liter
        resultUnit = gallonToLiter(initialUnit);

        //printing the results
        System.out.print("The volume in gallon is " + initialUnit + "
and in liter is " + resultUnit);
        break;
    }
    case 6 : {

        //taking user input
        System.out.print("Enter weight in pound : ");
        initialUnit = input.nextDouble();

        //calling the method liter to gallon to convert liter to gallon
        resultUnit = literToGallon(initialUnit);

        //printing the results
        System.out.print("The volume in liter is " + initialUnit + "
and in gallon is " + resultUnit);
        break;
    }
    default : {
        System.out.print("You have entered a wrong choice");
    }
}

```

```

    }

    //closing the input object
    input.close();
}
}

```

7. Write a program to take user input for the age of all 10 students in a class and check whether the student can vote depending on his/her age is greater or equal to 18.

Hint =>

- Create a class ***public class StudentVoteChecker*** and define a method ***public boolean canStudentVote(int age)*** which takes in age as a parameter and returns true or false
- Inside the method firstly validate the age for a negative number, if a negative return is false cannot vote. For valid age check for age is 18 or above return true; else return false;
- In the main function define an array of 10 integer elements, loop through the array by take user input for the student's age, call canStudentVote() and display the result

```

import java.util.Scanner;
public class StudentVoteChecker {

    //method to check whether a student can vote or not
    public static boolean canStudentVote(int age) {

        //validating input
        if(age > 0) {
            //returning true if student can vote else false
            if(age >= 18) {
                return true;
            } else {
                return false;
            }
        } else {
            return false;
        }
    }

    public static void main(String[] args) {

        //creating Scanner object for taking inputs

```

```

Scanner input = new Scanner(System.in);

//creating array ages to indicate ages of students
int[] ages = new int[10];

//taking ages of 10 students as user input
for(int i=0; i<10; i++) {

    System.out.print("Enter age of student " + (i+1) + " :
");

    ages[i] = input.nextInt();
}
//iterating through all students
for(int i=0; i<10; i++) {

    //calling the method canStudentVote and storing the
answer in variable result
    boolean result = canStudentVote(ages[i]);

    //printing whether the current student can vote or not
    if(result) {
        System.out.println("Student " + (i+1) + " can vote
whose age is " + ages[i]);
    } else {
        System.out.println("Student " + (i+1) + " cannot
vote whose age is " + ages[i]);
    }
}

//closing the input object
input.close();
}
}

```

8. Create a program to find the youngest friends among 3 Amar, Akbar and Anthony based on their ages and tallest among the friends based on their heights and display it

Hint =>

- a. Take user input for age and height for the 3 friends and store it in two arrays each to store the values for age and height of the 3 friends
- b. Write a Method to find the youngest of the 3 friends

c. Write a Method to find the tallest of the 3 friends

```
import java.util.Scanner;

public class HeightComparison3 {

    //method to find youngest of 3 friends
    public static int findYoungest(int[] ages) {

        // Variables to find the youngest
        int youngestIndex = 0;
        int minAge = ages[0];

        // Loop through the arrays to find the youngest
        for (int i = 1; i < 3; i++) {
            if (ages[i] < minAge) {
                // Check if the current friend is younger
                minAge = ages[i];
                youngestIndex = i;
            }
        }

        //returning the youngest index
        return youngestIndex;
    }

    //method to find tallest of 3 friends
    public static int findTallest(double[] heights) {

        // Variables to find the tallest
        int tallestIndex = 0;
        double maxHeight = heights[0];

        // Loop through the arrays to find the tallest
        for (int i = 1; i < 3; i++) {
            if (heights[i] > maxHeight) {
                // Check if the current friend is taller
                maxHeight = heights[i];
                tallestIndex = i;
            }
        }
    }
}
```

```

        //returning the youngest index
        return tallestIndex;
    }

    public static void main(String[] args) {
        // Initialize Scanner object with the name input
        Scanner input = new Scanner(System.in);

        // Arrays to store the ages and heights of the 3 friends
        String[] friends = {"Amar", "Akbar", "Anthony"};
        int[] ages = new int[3];
        double[] heights = new double[3];

        // Taking user input for ages and heights
        for (int i = 0; i < 3; i++) {
            System.out.print("Enter the age of " + friends[i] + ": ");
            ages[i] = input.nextInt();

            System.out.print("Enter the height (in cm) of " + friends[i] +
": ");
            heights[i] = input.nextDouble();
        }

        //creating variables to indicate youngest and tallest indexes
        int youngestIndex = 0;
        int tallestIndex = 0;

        //calling the findYoungest method to find youngest index
        youngestIndex = findYoungest(ages);

        //calling the findYoungest method to find youngest index
        tallestIndex = findTallest(heights);

        // Display the results
        System.out.println("\nThe youngest friend is " +
friends[youngestIndex] + " with age " + ages[youngestIndex] + " years.");
        System.out.println("The tallest friend is " + friends[tallestIndex]
+ " with height " + heights[tallestIndex] + " cm.");
    }
}

```

```
// Close the input object
input.close();
}
}
```

9. Write a program to take user input for 5 numbers and check whether a number is positive or negative. Further for positive numbers check if the number is even or odd. Finally compare the first and last elements of the array and display if they are equal, greater, or less

Hint =>

- Write a Method to Check whether the number is positive or negative
- Write a Method to check whether the number is even or odd
- Write a Method to compare two numbers and return 1 if number1 > number2 or 0 if both are equal or -1 if number1 < number2
- In the main program, Loop through the array using the length call the method **isPositive()** and if positive call method **isEven()** and print accordingly
- If the number is negative, print negative.
- Finally compare the first and last element of the array by calling the method **compare()** and display if they are equal, greater, or less

```
import java.util.Scanner;
public class ElementComparison3 {

    public static boolean isPositive(int number) {

        //checking number is positive, negative or zero
        if(number > 0) {
            return true;
        } else if(number < 0) {
            return false;
        } else {
            System.out.println("Number " + number + " is zero ");
            return false;
        }

    }

    public static boolean isEven(int number) {
        //checking whether the positive number is even or odd
        if(number % 2 == 0)
```

```

        {
            return true;
        } else{
            return false;
        }
    }
}

public static int compare(int numbers[]) {

    //comparing first and last element of array
    if( numbers[0] == numbers[4] )
    {
        return 0;
    } else if(numbers[0] > numbers[4]) {
        return 1;
    } else {
        return -1;
    }
}

public static void main(String []args) {

    //creating the scanner object
    Scanner input = new Scanner(System.in);

    //creating array numbers to indicate the numbers
    int numbers[] = new int[5];

    //taking 5 numbers as user input
    for(int i=0;i<5;i++){
        System.out.print("Enter number " + (i+1) + " : ");
        numbers[i] = input.nextInt();
    }

    //iterating through the numbers array
    for(int i=0;i<5;i++){

        //checking the number is positive or negative by calling
        //isPositive method
        if(isPositive(numbers[i])) {
            System.out.print("Number " + numbers[i] + " is
positive ");

```



```

//checking whether the positive number is even or
odd
        if(isEven(numbers[i])) {
            System.out.println( "and even.");
        } else{
            System.out.println("and odd.");
        }
    } else if(numbers[i]!=0) {
        System.out.println("Number " + numbers[i] + " is
negative ");
    }
}

//comparing first and last element of array
if(compare(numbers) == 0)
{
    System.out.println("Both numbers are equal");
} else if(compare(numbers) == 1) {
    System.out.println("First number is greater than the
last.");
} else {
    System.out.println("last number is greater than the first.");
}

//closing input object
input.close();
}
}

```

10. An organization took up the exercise to find the Body Mass Index (BMI) of all the persons in the team of 10 members. For this create a program to find the BMI and display the height, weight, BMI and status of each individual

Hint =>

- a. Take user input in double for the weight (in kg) of the person and height (in cm) for the person and store it in the corresponding 2D array of 10 rows and 3 columns. The First Column storing the weight, the second column storing the height in cm and the third column is the BMI

- Create a Method to find the BMI of every person and populate the array. Use the formula $BMI = \text{weight} / (\text{height} * \text{height})$. Note unit is kg/m^2 . For this convert cm to meter
- Create a Method to determine the BMI status using the logic shown in the figure below. and return the array of all the persons BMI Status.

BMI	Status
≤ 18.4	Underweight
18.5 - 24.9	Normal
25.0 - 39.9	Overweight
≥ 40.0	Obese

```
import java.util.Scanner;

public class BMICalculator3 {

    //method to calculate BMI
    public static double calculateBMI(double height, double weight) {

        //variable to indicate BMI
        double BMI;

        // Calculating BMI
        BMI = weight / (height * height);

        //returning the BMI
        return BMI;
    }

    //method to assign weight status
    public static String assignStatus(double BMI) {

        //variable to indicate status
        String status;

        //assigning weight status
        if (BMI <= 18.4) {
            status = "Underweight";
        } else if (BMI <= 24.9) {
            status = "Normal";
        } else if (BMI <= 39.9) {
```

```

        status = "Overweight";
    } else {
        status = "Obese";
    }

    //returning the status
    return status;
}

public static void main(String[] args) {
    Scanner input = new Scanner(System.in);

    // creating Array detail to store height (1st column), weight (2nd
    column), BMI (3rd column)
    double[][] details = new double[10][3];

    //creating Array to store status
    String[] statuses = new String[10];

    // Taking input for height and weight
    for (int i = 0; i < 10; i++) {
        System.out.println("Person " + (i + 1) + ":");
        System.out.print("Enter height (meters): ");
        details[i][0] = input.nextDouble();
        System.out.print("Enter weight (kg): ");
        details[i][1] = input.nextDouble();

        // Validating inputs
        if (details[i][0] <= 0 || details[i][1] <= 0) {
            System.out.println("Invalid input. Try again.");
            i--; // Retrying for the current person
            continue;
        }
    }

    //iterating through the details to calculate BMI's and
    assigning statuses
    for(int i=0; i<10; i++) {

        //calling the method calculateBMI to calculate BMI

```

```

        details[i][2] = calculateBMI(details[i][0],
details[i][1]);

        //calling the method assignStatus to assign status
        statuses[i] = assignStatus(details[i][2]);
    }

    // Displaying results
    System.out.println("\nResults:");
    for (int i = 0; i < 10; i++) {
        System.out.println("Person " + (i + 1) + ":");
        System.out.println("Height: " + details[i][0] + " meters");
        System.out.println("Weight: " + details[i][1] + " kg");
        System.out.println("BMI: " + details[i][2]);
        System.out.println("Status: " + statuses[i]);
    }

    //closing the input object
    input.close();
}
}

```

11. Write a program Quadratic to find the roots of the equation $ax^2 + bx + c$. Use Math functions **Math.pow()** and **Math.sqrt()**

Hint =>

- Take a, b, and c as input values to find the roots of x.
- The roots are computed using the following formulae

$$\text{delta} = b^2 + 4 * a * c$$

If delta is positive the find the two roots using formulae

$$\text{root1 of } x = (-b + \sqrt{\text{delta}})/(2 * a)$$

$$\text{root1 of } x = (-b - \sqrt{\text{delta}})/(2 * a)$$

If delta is zero then there is only one root of x

$$\text{root of } x = -b/(2 * a)$$

If delta is negative return empty array or nothing

- Write a Method to find find the roots of a quadratic equation and return the roots

```
import java.util.Scanner;
import java.lang.Math;
public class QuadraticRoots {

    //method to calculate roots of quadratic equation
    public static double[] findRoots(double a, double b, double c) {
        //creating variable delta to indicate delta
        double delta;

        //calculating delta
        delta = Math.pow(b,2) - 4 * a * c;

        //checking whether delta is positive, negative or zero
        if(delta > 0) {
            //calculating two roots for positive delta
            double root1 = (-b + Math.sqrt(delta))/(2 * a);
            double root2 = (-b - Math.sqrt(delta))/(2 * a);

            //storing roots in array
            double roots[] = {root1, root2};

            //returning the roots
            return roots;
        } else if(delta == 0) {

            //calculating one root for zero delta
            double root = -b / (2 * a);

            //storing roots in array
            double roots[] = {root};

            //returning the roots
            return roots;
        } else {

            //storing roots as empty array for negative delta
            double roots[] = {};
            //returning the
            return roots;
        }
    }
}
```

```

public static void main(String []args) {

    //creating scanner object to take user input
    Scanner input = new Scanner(System.in);

    //creating variables a, b and c
    double a, b, c;

    //taking user input for a, b, and c
    System.out.print("Enter the value of a : ");
    a = input.nextDouble();
    System.out.print("Enter the value of b : ");
    b = input.nextDouble();
    System.out.print("Enter the value of c : ");
    c = input.nextDouble();

    /*calling the QuadraticRoots method to calculate roots and
storing
roots in array roots*/
    double roots[] = findRoots(a, b, c);

    //printing the roots
    if(roots.length == 0) {
        System.out.print("There is no roots.");
    } else if (roots.length == 1) {
        System.out.print("There is only one root i.e. " +
roots[0]);
    } else {
        System.out.print("There are two roots " + roots[0] + "
and " + roots[1]);
    }
    //closing the input object
    input.close();
}
}

```

12. Write a program that generates five 4 digit random values and then finds their average value, and their minimum and maximum value. Use Math.random(), Math.min(), and Math.max().

Hint =>

- a. Write a method that generates array of 4 digit random numbers given the size as a parameter as shown in the method signature

public int[] generate4DigitRandomArray(int size)

- b. Write a method to find average, min and max value of an array

public double[] findAverageMinMax(int[] numbers)

```
import java.util.Scanner;
import java.lang.Math;
public class RandomMaxMinAverage {

    //method to generate random numbers
    public static int[] generate4DigitRandomArray(int size) {

        //creating array numbers to store random numbers
        int numbers[] = new int[size];

        //generating random numbers and storing into array
        for(int i=0; i<size; i++) {
            numbers[i] = (int) (Math.random() * 9000) + 1000;
        }
        //returning the array
        return numbers;
    }

    //method to calculate maximum, minimum and average of numbers
    public static double[] findAverageMinMax(int[] numbers) {

        //creating array maxMinAverage to store maximum, minimum and
        //average of numbers
        double maxMinAverage[] = new double[3];

        //calculating maximum of numbers
        maxMinAverage[0] = numbers[0];
        for(int i=1; i<numbers.length; i++) {
            if(numbers[i]>maxMinAverage[0]) {
                maxMinAverage[0] = numbers[i];
            }
        }
    }
}
```

```
//calculating minimum of numbers
maxMinAverage[1] = numbers[0];
for(int i=1; i<numbers.length; i++) {
    if(numbers[i]<maxMinAverage[1]) {
        maxMinAverage[1] = numbers[i];
    }
}

//creating sum variable to store sum of number
int sum = 0;

//calculating sum
for(int i=0; i<numbers.length; i++) {
    sum += numbers[i];
}

//calculating average of numbers
maxMinAverage[2] = sum/numbers.length;

//returning the array
return maxMinAverage;
}

public static void main(String []args) {

    //creating scanner object to take user input
    Scanner input = new Scanner(System.in);

    //creating variables size to indicate size of array
    int size;

    //taking user input for size
    System.out.print("Enter the number of elements : ");
    size = input.nextInt();

    //creating array numbers to store random numbers
    int numbers[] = new int[size];

    //creating array maxMinAverage to store maximum, minimum and
    average of numbers
    double maxMinAverage[] = new double[3];
```



```

        /*calling the generate4DigitRandomArray method to generate
random numbers*/
        numbers = generate4DigitRandomArray(size);

        //calling the findAverageMinMax method to calculate max, min
and average
        maxMinAverage = findAverageMinMax(numbers);

        //printing numbers
        System.out.print("numbers are :");
        for(int i=0; i<size; i++) {
            System.out.println(" " + numbers[i]);
        }

        //printing details
        System.out.println("Maximum of numbers is " + (int)
maxMinAverage[0]);
        System.out.println("Minimum of numbers is " + (int)
maxMinAverage[1]);
        System.out.println("average of numbers is " +
maxMinAverage[2]);

        //closing the input object
        input.close();
    }
}

```